ROTATION: WELL-CHILD CLINIC

Chapter 4 — Well-Child Clinic

The central practical site for health supervision: the recommended visit schedule, what happens at each visit, age-specific anticipatory guidance for every stage (0-20 yr), immunization administration in the clinic, and a milestone quick reference with red flags.

Age range: Birth – 20 years

4.1 Site Profile — Role of the Well-Child Clinic

The well-child clinic is where positive health is promoted in normal, healthy children. Visits focus primarily on **anticipatory guidance** — teaching what to expect next and how to prevent harm before the child reaches the susceptible age — ideally beginning during pregnancy.

- **Goal:** periodic screening and the prevention of disease. The AAP recommends periodic check-ups for **all** infants and children.
- **Nurse's role:** carry out the recommended periodic screenings and the current immunization schedule; educate the family at every visit.
- **Anticipatory guidance:** one third of all childhood injuries occur in the home — most are predictable and preventable when teaching precedes the susceptible age.
- **Cultural respect:** explore the family's cultural practices around growth and development; most are harmless and can be supported. Never assume practices from skin color, accent, or name.

FOUR SKILLS USED AT EVERY VISIT

- **History:** interval history — feeding, elimination, sleep, behavior, past illnesses, allergies, medications, immunization record.
- **Measurements & plotting:** length/height, weight, head circumference, weight-for-length or BMI, and increasingly blood pressure.
- **Screening:** vision, hearing, and developmental/behavioral screening and surveillance.
- **Guidance & procedures:** physical examination, immunizations and other procedures, then anticipatory guidance and parent education.

4.2 Health Supervision Visit Schedule

The current recommendations for periodic health visits start in the newborn period and continue through adolescence. For the guide's 0-20-year population the sequence runs: a newborn visit, a 3-5-day visit, then visits by 1 month and at 2, 4, 6, 9, 12, 15, 18, 24, and 30 months; from 3 years onward the visits become annual, with a closing visit at 20 years. The sequence is charted in Figure 4.1.

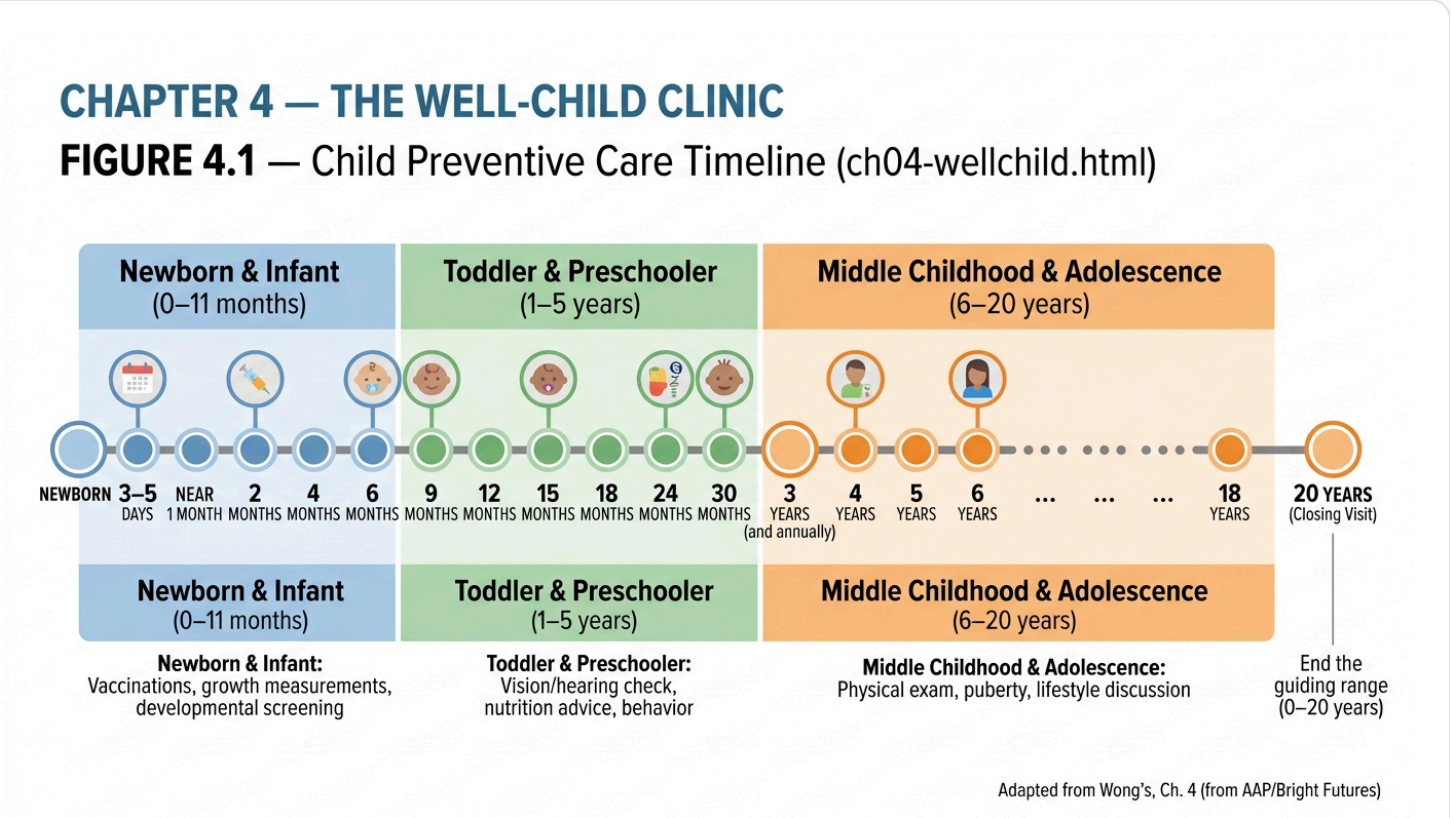


Figure 4.1 Health-supervision visit sequence for the 0-20-year population.

Service category	When it is done
History (interval history, health history)	At every visit
Measurements — length/height, weight, head circumference, weight-for-length or BMI, blood pressure	Length/height/weight/HC at every visit in the first years; BMI and blood pressure from ~3 yr; **BP annually**

Vision screening	USPSTF recommends screening for amblyopia and risk factors in **all children 3-5 years of age**; elective photoscreening/autorefraction possible from 6 months-3 years (early amblyopia detection)
Hearing screening	Pure-tone audiometry screening at 0.5, 1, 2, and 4 kHz — fail when tones at 20 dB HL are not heard; hearing concerns asked at **every** visit; universal newborn hearing screening already done (Ch. 3)
Developmental & behavioral	Surveillance at every visit; formal developmental screening and autism screening at the ages mapped in Fig. 4-7 (Ch. 2 covers Denver II, ASQ, PEDS)
Psychosocial & behavioral (alcohol/drug later)	From childhood; alcohol/drug discussion in adolescence (Ch. 8)
Physical examination	At every preventive visit
Procedures — newborn metabolic screen, immunizations, hematocrit, lead, TB, dyslipidemia, STI, oral health	Per the visit-schedule figure and Section 4.5
Anticipatory guidance	At every visit — the central content of this chapter (Section 4.4)

4.3 The Well-Child Visit, Step by Step

A preventive visit follows a fixed order — history first, measurements and plotting, sensory and developmental screening, the physical examination, then procedures and parental guidance. The workflow below is the same for every age; only the content changes. Figure 4.2 charts this sequence as a diagram.

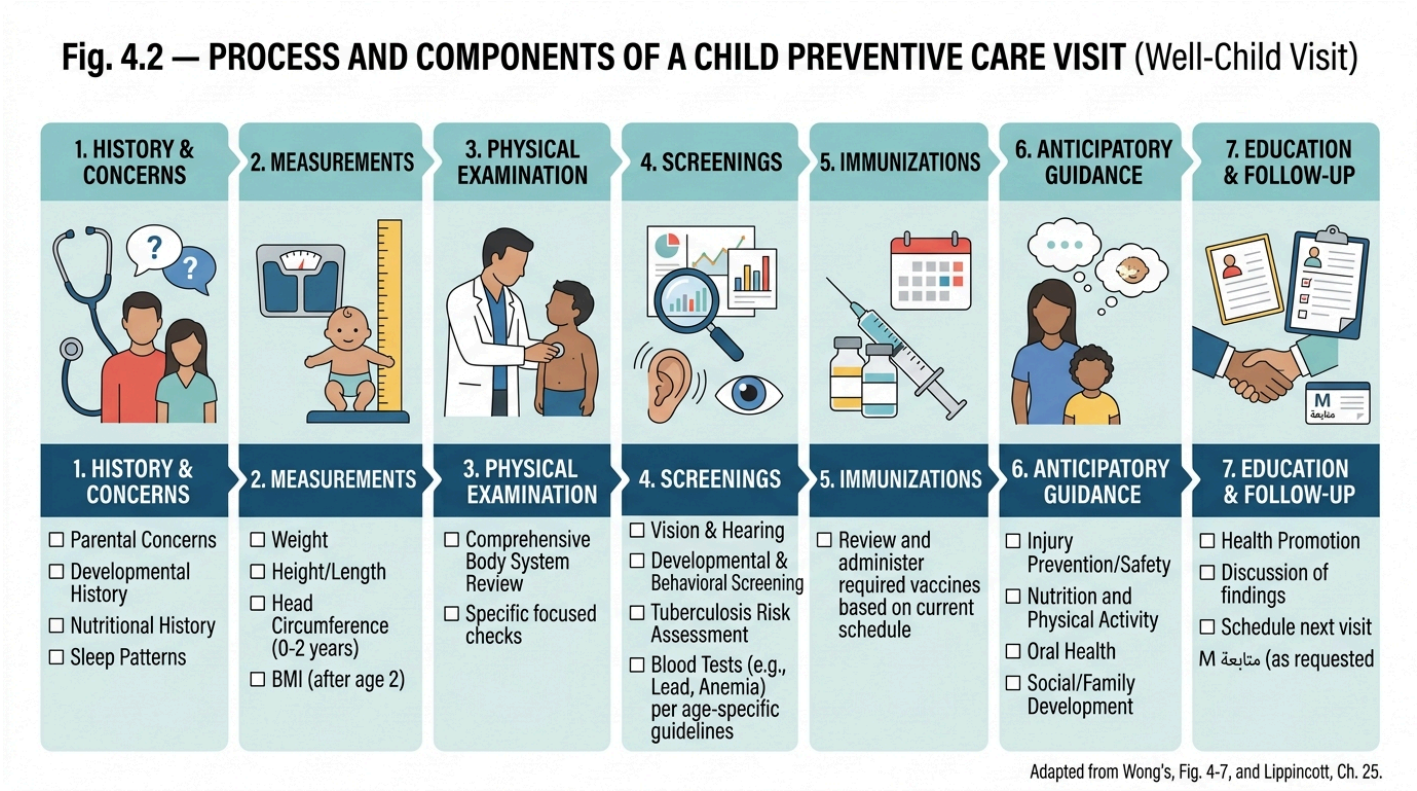


Figure 4.2 Well-child visit workflow, from history to anticipatory guidance.

History essentials

Component	Detail
Birth history	(1) Mother's health during pregnancy, (2) labor and delivery, (3) the infant's condition immediately after birth
Past illnesses	Colds, earaches, measles, rubella, chickenpox, mumps, pertussis, diphtheria, tuberculosis, scarlet fever, strep throat, recurrent ear infections, GERD, tonsillitis, allergic manifestations
Injuries / operations	Past injuries needing medical care, operations, hospitalizations — with dates
Allergies	Hay fever, asthma, unusual drug/food/latex reactions, contact agents
Current medications	Vitamins, antipyretics (especially aspirin), antibiotics, antihistamines, decongestants, herbs/homeopathics; record name, dose, schedule, duration, reason; ask parents to bring the containers

Immunization record	Essential and often unknown to parents — the most reliable source is the hospital/clinic/private-practitioner record. Record: disease, number of injections, dosage, ages given, and any post-immunization reaction
Growth history	Weights ≈ 6 months, 1, 2, and 5 years; lengths ≈ 1 and 4 years; dentition (onset, number, symptoms)
Milestones	Head-holding, sitting alone, walking, first words; school grade and performance; best friend; peer interactions

Measurement & plotting rules

- **Length:** supine on a **length board** until 24 months (or 36 months via the birth-36-month chart); a tape measure or wall chart is **never used**. Measure at least twice (ideally 3 times) with values agreeing within 0.5 cm; use the mean.
- **Height:** standing with a **stadiometer**; wall charts and flip-up bars are not used.
- **Weight:** nude for the birth-36-month charts; balance-beam scale to the nearest 10 g (infant) or 100 g (child).
- **Head circumference:** measured and plotted through **36 months**; head and chest circumferences are roughly equal at about 1-2 years.
- **Charts:** WHO growth standards 0-2 years (breastfed reference); CDC charts for 2-20 years with BMI-for-age.
- **Interpretation:** a normal, well-nourished child plots between the **5th and 95th percentiles**; below the 5th or above the 95th suggests deficiency or excess. A child at the 5th percentile can be as normal as one above the 95th — the most reliable evaluation is the **trend over time**.
- **Concern:** widely disparate percentiles (e.g., height 10th + weight 90th), failure to follow expected growth velocity, or a sudden change in a previously steady pattern.
- **Technique:** when growth is a concern, take serial measurements at the **same time of day** (afternoon height is less than morning height).

Figure 4.3 shows the percentile bands on a growth chart: adequate nutrition falls between the 5th and 95th percentiles; for BMI-for-age the 85th-95th band marks risk and the 95th marks overweight. These rules are given in the measurement tips above.

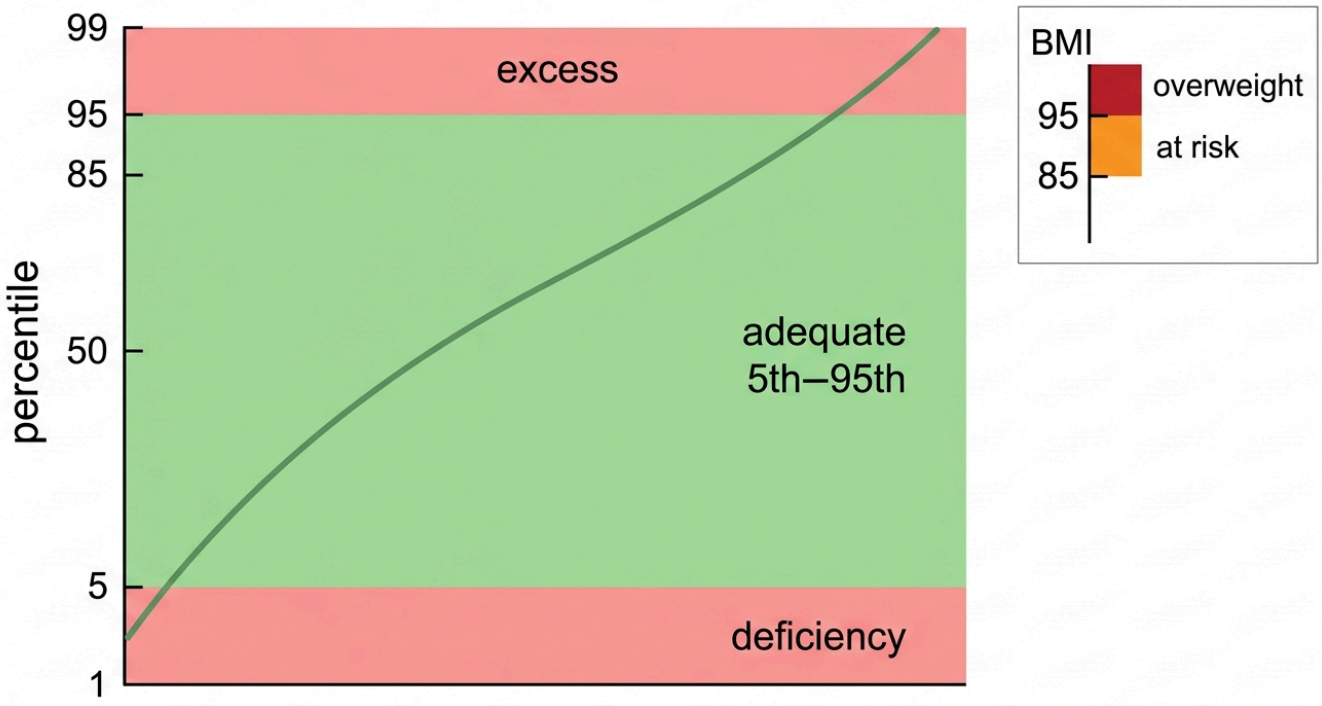


Figure 4.3 Growth-chart percentile bands and their interpretive thresholds.

4.4 Age-Specific Health Supervision

Anticipatory guidance is matched to each developmental stage. The five blocks below give the essential teaching points for each age.

Infant — 0-12 months

Feeding topic	Guidance
Solids	Do not add solids before **4-6 months** (earlier: sixfold increase in obesity odds in formula-fed infants); transition period ≈ 4-6 months. Iron-fortified infant cereal first (rice — easily digested, low allergic potential); one new food at a time, every **5-7 days** (one source says 4-7, another 3-5 days); raw seeds/nuts/vegetables and grapes are aspiration hazards; cracked zwieback ~6 months; junior/finger foods 8-9 months; well-cooked table foods at 1 year
Milk	Milk remains the main calorie source until after **12 months**; whole cow's milk **not** before 12 months (iron/zinc/vitamin C deficient, high renal solute load); skim/low-fat also unacceptable

Avoid	Honey in the first 12 months (infant botulism); propping a bottle or bottle-in-bed; juice in bottles, especially <6 months (enamel erosion, early childhood caries)
Supplements	Vitamin D **400 IU daily** for all infants (those on <1 L/day of fortified formula included); iron 1 mg/kg/day if exclusively breastfed >4 months until iron-fortified foods start
Juice	Not required under 6 months and of no demonstrable benefit; only 100% juice; max 4-6 oz/day (1-6 yr); avoid apple, pear, prune, sweet cherry, peach, grape (fructose/sorbitol → cramps/diarrhea); give sugared liquids in a cup
Cup milestones	Lips to cup rim 4-5 months; drinks from a cup with help 7-9 months; drinks from a cup (with spilling) 9-12 months
Weaning	Gradual, one feeding at a time; the nighttime feeding is discontinued last; readiness signals appear in the second half of the first year

- **Sleep:** place the infant to sleep on the **back at all times**; firm mattress, no pillows or loose blankets; crib slats spacing ≤6 cm; no drop-side cribs. Most infants sleep 9-11 hours nightly at 3-4 months (~15 hr/day total); ~11.7 hr nightly by 1 year.
- **Pacifier:** AAP supports **limited** use in the first 6 months (analgesia + SIDS protection), weaning in the second 6 months; one-piece construction with a loop handle; never on a string around the neck.
- **Car seats:** all infants and toddlers ride **rear-facing for as long as possible — until they reach the highest weight or height allowed by the car-seat manufacturer** (most convertible seats allow rear-facing past age 2; older editions: "until 12 months AND 20 lb" — report the current recommendation); rear-facing seat **never** in a front seat with a passenger air bag; back seat is safest; ~52% of newborn seats are misused at hospital discharge.
- **Dental:** first teeth (lower central incisors) 6-10 months, average 8 months; teeth ≈ age in months minus 6; clean gums/teeth with a damp cloth; fluoridated toothpaste "smear" for children <2 years; **oral examination by 6 months and a dental home by 1 year**; fluoride supplement from 6 months if the water is inadequate; watch the infant for earaches/circumoral cyanosis.
- **Teething:** fever >39°C, vomiting, or diarrhea means illness — not teething; cold soothes; no teething powders, aspirin, or liquor on the gums; no amber necklaces.
- **Ears & language:** cooing / simple vowel sounds and laughing aloud appear about 4-5 months; babbling with inflection 7-10 months. Warning signs: no sounds or babbling by 4 months; not laughing or squealing by 6 months.
- **Play:** solitary; suggested toys 0-6 months (mobiles, unbreakable mirror, music box, rattles) and 6-12 months (blocks, nested cups, picture books, ball, cup and spoon, wood puzzles, push-pull toys); restrict TV and recorded video in children <2 years.

- **Discipline:** infants cannot grasp cause-and-effect of danger — childproof the environment; begin with a negative voice and stern eye contact, then time-out (play yard preferred over a chair); there is no evidence that meeting needs spoils a child.

INFANT RED FLAGS

- No smile at people at **3 months**; refuses cuddling; no interest in peek-a-boo at **8 months** (social-emotional)
- No head support 3-4 months; no sitting with assistance by 6 months; no crawling by 12 months; no supported standing by 12 months
- Not pulling to stand by 11-12 months → evaluate for developmental hip dysplasia
- Strabismus / lack of binocularity can cause permanent blindness if undetected

Toddler — 1-3 years

- **Nutrition:** 1000-1400 kcal/day (moderately active boy, 2-3 yr, 2010 Dietary Guidelines); growth slows 12-18 months and physiologic anorexia (decreased appetite) appears around 18 months — a normal, not alarming, event; 3 meals + 2 snacks with small portions; milk 2-3 servings (24-28 oz/day — more than a quart limits solids and leads to iron/nutrient deficiency); low-fat (2%) milk after age 2; juice capped at 4-6 oz/day of 100% juice.
- **Weaning:** drink from a cup at the first birthday; weaning from the bottle is complete by 15 months (the 12-15-month window); never a bedtime bottle — ends "nursing bottle caries" (early childhood caries); dental care: water-only brushing before age 2, then a pea-sized amount of fluoridated toothpaste beginning at 2 years.
- **Nutrition numbers:** iron EAR 7 mg/day (1-3 yr); calcium AI 500 mg/day (1-3 yr).
- **Negativism:** it is a necessary assertion of control, not stubbornness — reduce the opportunities for "no" (offer two acceptable choices); if refused, the parent calmly decides; ritualism (need for sameness) is comforting and helps through hospitalization.
- **Tantrums:** common when ill, hungry, frustrated, tired (worst near bedtime); three quarters last ≤5 minutes; manage by ignoring the behavior unless injurious, staying present, comforting once it subsides, and praising non-tantrum behavior. **Red flags:** tantrums past 5 years, lasting over 15 minutes, or more than 5 times a day → possible serious problem. Time-outs work well from ~18 months.
- **Toilet training:** readiness has five markers (bladder, bowel, cognitive, motor, psychologic) — sphincter control usually by 22-30 months, dry ≥2 hours, waking from the nap dry, regular bowel movements, sits/walks/squats, removes clothing; mental readiness (recognizes the urge,

communicates, imitates); psychologic (wishes to please, sits 5-8 min); and parental readiness (no family stress from move, divorce, new sibling). Average initiation ~20.6 months; girls 2-2.5 months before boys; no benefit of intensive training before 27 months; bowel training precedes bladder; nighttime dryness takes months-years (bedwetting normal in girls ≤4, boys ≤5; no night dryness by 6 years → intervene). Regression is a normal temporary setback.

- **Sleep:** ~11-12 hours with one nap; bedtime resistance and fear of monsters are common later — regular routine, transitional object, snack, TV out of the room.
- **Media:** television before age 2 decreases language skills; total quality media ≤1 hour/day for toddlers.
- **Play:** parallel play (alongside, not with others) with imitation; the best "toy" is adult-child interaction.
- **Safety:** unintentional injury is the leading cause of death ages 1-19 (~37%); rear-facing car seat for as long as possible up to the manufacturer's weight/height limit, then a forward-facing seat with harness, then a belt-positioning booster until 145 cm (4 ft 9 in) or 8-12 years; gates at both stair ends; waterproofing — the highest drowning rate occurs in 1- to 2-year-olds, so constant supervision is needed in and around all water; water heater ≤49°C; poisoning only locked cabinets protect (cosmetics, analgesics, cleaners top exposures); hot-car hyperthermia deaths are preventable.

Preschooler — 3-6 years

- **Development:** Erikson initiative vs guilt; the superego/conscience develops; magical thinking makes words literal — say "that was a bad thing to do," not "you are bad"; identification with the same-sex parent resolves the Oedipal stage.
- **Fears:** the greatest number and variety of fears appear in the preschool years (dark, being left alone, animals, ghosts, annihilation); most resolve by 5-6 years. Management: night-light, desensitization/modeling, relaxation, brave self-talk. **Refer** severe fears that disrupt family life.
- **Imaginary friends** usually appear 2½-3 years and last until school entry (more common in firstborns and girls) — a sign of health; parents may acknowledge the friend but the child stays responsible for behavior.
- **Play:** associative play (similar activity, no rigid rules); dramatic/imitative play is central; 4-5-year-olds reproduce significant-adult behavior.
- **Language:** vocabulary grows from ~300 words at 2 years to >2100 by the end of 5 years.
- **School readiness:** no single readiness indicator; social maturation (especially attention span) matters as much as academics; school begins between ages 5 and 6.
- **Nutrition:** ~90 kcal/kg ≈ 1800 kcal/day; fluids ~100 ml/kg/day; protein 13-19 g/day; fiber = age + 5 g; saturated fat <10% of calories; calcium 1000 mg/day (4-8 yr); juice/sweetened drinks ≤4-6 oz/day;

follow the **5-2-1-0** framework (5 fruits/vegetables, ≤2 hours of screen, 0 sugar-sweetened drinks, 1 hour activity); preschoolers eat about half an adult portion.

- **Sleep & dental:** ~12 hours nightly with infrequent naps; night waking is common. Primary dentition is complete by the start of the preschool years; brushing still needs adult assistance and flossing is a parent's task; continue professional care and fluoride.
- **Discipline:** verbal limit-setting works better than with toddlers — the child obeys danger warnings; allowing questions and disagreement fosters independence; time-out and solitary play stop aggression; spanking teaches aggression "as a lesson"; encourage verbal expression of feelings.
- **Safety:** less prone to falls and more aware of danger; poisoning still dangerous (mouthing has ceased); pedestrian-vehicle injuries increase; teaching replaces pure protection — parents model behavior (bicycle helmets); electrical-outlet caps can be discontinued with an explanation.
- **Red flags:** severe fears disrupting family life; extreme aggression (needs professional treatment); stuttering with no improvement over >6 months → speech evaluation; injury-prevention education for 4½-year-old risk takers.

School-age — 6-12 years

- **Growth:** ~5 cm/yr; weight gain 2-3 kg/yr; average 6-year-old ≈116 cm/21 kg; average 12-year-old ≈150 cm/40 kg; girls surpass boys in height and weight near the end. Prepubescence is the ~2-year period preceding puberty; first physiologic signs about 9 years (girls). All primary teeth are shed in this span — the first tooth (central mandibular incisors) is lost at about 6 years ("age of the loose tooth"); the 6-year molar erupts behind the deciduous molars; 60% of children have dental caries by age 5.
- **Psychosocial:** Erikson industry vs inferiority — competence through skills; peers become central (same-sex "best friends", clubs with codes and conformity); bullying affects ~25% of elementary students as bully or victim, and cyberbullying can be more harmful (anonymous, wide audience); teach reporting and family conversation. Physical maturity ≠ emotional maturity.
- **Cognitive:** concrete operations 7-11 years; conservation of number 5-6, liquid/mass/length 6-7, weight 9-10, volume 9-12; reading is the most significant school skill; homework: quiet, well-lit area, no TV until done, explain rather than answer.
- **Nutrition:** caloric needs are lower than preschool and lower than the adolescent growth spurt; children eat what the family eats; encourage fruit, nuts, and wholesome snacks; obesity prevalence is stable at ~17% and higher in rural/socioeconomically distressed families; school-lunch programs must supply 1/3 of the daily RDA.
- **Safety:** motor-vehicle crashes are the most common cause of severe accidental injury/death (as passenger or pedestrian); children <13 years ride in the rear, never the front passenger seat with an

air bag; over 18 kg (40 lb) use a forward-facing convertible seat if it fits, then a booster until the lap-shoulder belt fits; no one rides in a pickup-truck bed; ATVs are not recommended under 16 years; bicycle helmets cut head injury ~45%, brain injury ~33%, facial injury ~27%, fatal injury ~29% — helmet every ride, the bike must fit; trampolines: single jumper, no somersaults, ≥6 years; skateboards <5 years never, 6-10 only with close supervision and full padding; participation physical exams ≥every 2 years for sports; firearms locked.

- **Sleep & dental:** sleep falls from ~11.5 hours at 5 years to ~9 hours at 11; up to 50% of children have nightmares; sleepwalking is common (3-5% frequent) in the first 3-4 hours of sleep. Children may need help with brushing until ~7-10 years — parents monitor and floss at least daily; brush 2-3×/day for 2-3 minutes with fluoridated toothpaste; dental examinations every 6 months; dental sealants are recommended for all school-age children.
- **Discipline:** no corporal punishment — reasoning is effective; time-out for acute behavior; advanced strategies include withholding privileges, recompense, penalties, and contracting; problem-solving is the best limit-setting approach; discipline works best in a positive, supportive relationship.
- **Guidance box highlights:** age 6 — ravenous appetite, bicycle safety, privacy (separate bedroom), peer interaction; ages 7-10 — allergies may appear, minor injuries increase, growth of peer involvement, independence with limits; ages 11-12 — prepare for pubescent body changes, accurate sex education, allow regression while supporting growing up.
- **Stress signals (explore further):** stomach pains/headache, sleep problems, bed-wetting, eating-habit changes, aggressive/stubborn behavior, reluctance to participate, regression (e.g., thumb sucking).

Adolescent — 12-20 years

The full adolescent assessment (confidentiality, Tanner staging, HEEADSSS interview, risk-behavior screening, annual visits) is built in **Chapter 8**. Guidance points that belong to every clinic visit here:

- **Annual influenza vaccination** (live-attenuated or inactivated) is recommended for all adolescents.
- **Tdap:** recommended at 11-18 years for those who completed the childhood DTaP/DTP series and have not received a tetanus booster; may be given before the 5-year Td interval to close the pertussis gap.
- **Meningococcal (MenACWY):** at 11-12 years with a booster before age 16; a single dose at 13-18 years if never vaccinated.
- **HPV:** the 9-valent vaccine (Gardasil 9) from a minimum age of 9 years, routine at 11-12 years; **two doses** if the series starts at 9-14 years, **three doses** for a start at 15 years or later or in immunocompromised persons.

- **Varicella:** for those without a history of infection — two doses ≥4 weeks apart for ages ≥13 years. **Hepatitis B:** three-dose series for anyone not previously covered. **Hepatitis A** in targeted areas or when immunity is desired.
- **Confidentiality:** offers of confidential services are critical for sensitive topics (sexual behavior, substance use); many teenagers avoid care if they believe their parents will be told.
- **HEEADSSS** interview (Ch. 8): Home / Education / Eating / Activities / Drugs / Sexuality / Suicide-depression / Safety.

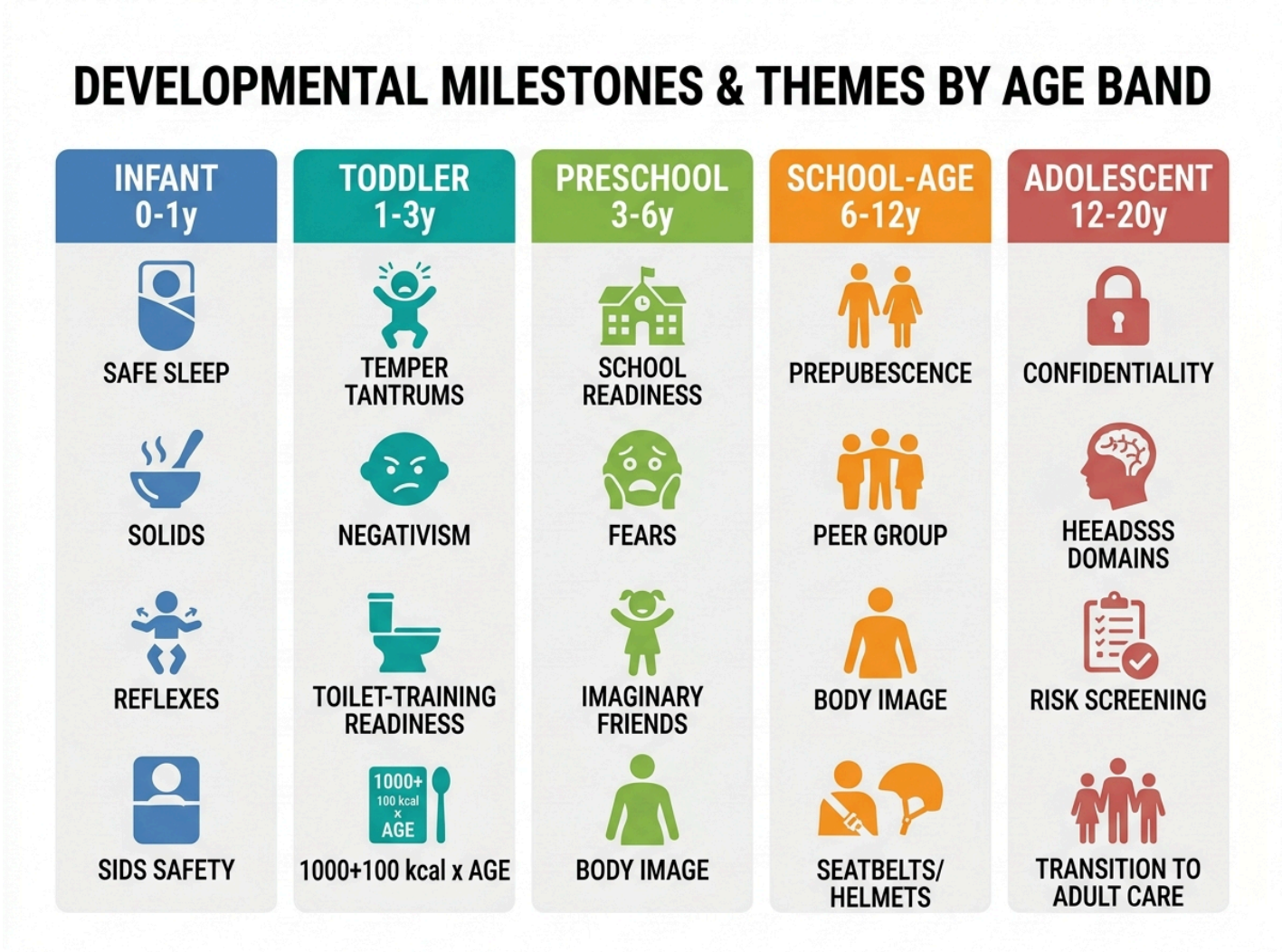


Figure 4.4 Anticipatory-guidance themes across the five age blocks: infant, toddler, preschooler, school-age, and adolescent (full adolescent assessment in Chapter 8).

4.5 Immunization Administration in the Clinic

Immunization is a central component of every preventive visit. The steps below cover the **record, the history, and the age-specific immunizations** used in the clinic. Whenever a vaccine is due,

use the official printed schedule, which is the definitive reference for routine infant and toddler doses.

Step	Detail
Immunization record	Essential at every visit; parents are often unaware of exact names and dates — the most reliable record is the hospital/clinic/private-practitioner record. For each immunization and booster record: disease, number of injections, dosage, ages given, and any post-immunization reaction
History before vaccinating	Recent immune globulin or blood transfusion necessitates a **delay in live vaccines**; ask about prior horse/foreign serum and anaphylactic reactions to neomycin, eggs, or vaccine components
Stress immunization needs	Anticipatory guidance at infant visits must stress the need for immunization
Newborn doses (Ch. 3)	Vitamin K IM 0.5-1.0 mg vastus lateralis; hepatitis B dose 1 (birth-2 months) — see Chapter 3
Adolescent catch-up/boosters	Tdap at 11-18 yr for those not yet boosted; MenACWY at 11-12 yr with booster before 16; HPV series from minimum age 9; annual influenza; varicella if no history (2 doses ≥4 wk apart, ≥13 yr); hepatitis B series; hepatitis A in targeted areas

4.6 Milestone Quick Reference & Red Flags (0-20 yr)

The tables below make a clinic quick-reference of the milestone ages; the Denver II remains the definitive screening measure.

Age / period	Milestone
~2 months	First real smile at people
3-4 months	Head support expected (no head support = red flag)
5-8 months	Sits with back upright (supported) 5 mo; tripod sit 6 mo; sits alone with hand support 7 mo; sits unsupported 8 mo
8 months	Interest in peek-a-boo expected (missing = warning sign); creeping/crawling emerges

11-12 months	Pulls to stand (missing at 11-12 mo → hip-dysplasia evaluation); walks independently ~12 mo
12 months	2-3 recognizable words with meaning; birth weight tripled; length +50%; HC +33%
18-24 months	Voluntary sphincter control; readiness for toilet training typically stands (average start 20.6 mo; daytime control often by 24 mo)
2-5 years	Vocabulary ~300 words (2 yr) grows to >2100 words (5 yr); imaginary friends 2½-3 yr until school entry; most fears resolve by 5-6 yr
5-6 years	School readiness social maturation; school begins 5-6 yr; first deciduous tooth (central mandibular incisors) shed ~6 yr
7-11 years	Concrete operations; conservation of number 5-6, liquid/mass/length 6-7, weight 9-10, volume 9-12
11-14 years	Peak height velocity ≈12 yr (girls) and ≈14 yr (boys); menarche mean ≈12.55 yr (Caucasian), 12.06 (African-American), 12.25 (Mexican-American)
12-20 years	Adolescent care — see Chapter 8 (Tanner, HEEADSSS, risk behaviors)

Age / period	Red flag — investigate or refer
0-3 months	No smile at people at 3 months; child refuses to cuddle; no head support by 3-4 months
4-8 months	No sitting with assistance at 6 months; no interest in peek-a-boo at 8 months; strabismus / no binocularity (can cause permanent blindness)
9-12 months	No crawling or supported standing by 12 months; not pulling to stand by 11-12 months; no single words with meaning at 12 months (ma-ma, da-da)
Toddler (1-3 yr)	Tantrums over 5 years of age, lasting over 15 minutes, or more than 5 times/day; bedwetting not improving — no night dryness by age 6 suggests the need for intervention
Preschool (3-6 yr)	Severe fears disrupting family life; extreme aggression; stuttering with no improvement over more than 6 months
School-age (6-12 yr)	Stress signals: stomach pains/headache, sleep problems, bed-wetting, eating-habit changes, aggressive/stubborn behavior, reluctance to participate, regression (thumb sucking); exposure to bullying/cyberbullying

Adolescent (12-20 yr)	See Chapter 8 — substance use, depression/suicidality, risky sexual behavior, safety (HEEADSSS positive screens)

Chapter 4 Learning Check

Self-assessment questions (answer in your notebook, then open)

End of Chapter 4.